MAT 2040 Linear Algebra

Assignment 5

Note: Submission of this assignment is NOT required.

Question 1. Suppose $A = \begin{bmatrix} 1 & 2 & 3 & 5 & 0 & 2 & 4 \\ 2 & 1 & 3 & 4 & 7 & 8 & 9 \\ 1 & 1 & 2 & 3 & 2 & 2 & 5 \end{bmatrix}$.

(a) Find a basis for $\text{Col}(A)$.

(b) What is the dimension of $\text{Row}(A)$?

(c) Find a basis for $\text{Null}(A)$ and find the dimension of $\text{Null}(A)$.

Solution

(a) We do RREF can get the matrix:

$$\begin{bmatrix} 1 & 2 & 3 & 5 & 0 & 2 & 4 \\ 2 & 1 & 3 & 4 & 7 & 8 & 9 \\ 1 & 1 & 2 & 3 & 2 & 2 & 5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & -14 & 14 \\ 0 & 1 & 1 & 2 & 0 & 8 & -5 \\ 0 & 0 & 0 & 0 & 1 & 4 & -2 \end{bmatrix}$$

So the pivot columns of B are $\{1, 2, 5\}$. Hence the basis for $\text{Col}(A)$ is $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 7 \\ 2 \end{bmatrix} \right\}$

(b) From the RREF matrix of (a), we can write:

$$\begin{cases} x_1 = -x_3 - x_4 + 14x_6 - 14x_7 \\ x_2 = -x_3 - 2x_4 - 8x_6 + 5x_7 \\ x_5 = -4x_6 + 2x_7 \end{cases}$$

So the basis of Null(A) is

$$\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 14 \\ -8 \\ 0 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -14 \\ 5 \\ 0 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Hence the dimension of Null(A) is 4.

Question 2. For the following matrix, find a basis for the row space, a basis for the column space, and a basis for the null space.

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 4 & 7 & 8 \end{bmatrix}$$

Solution

The reduced row echelon form of the matrix is

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus (1, 0, 2) and (0, 1, 0) form a basis for the row space. The first and second columns of the original matrix form a basis for the column space:

$$\mathbf{a}_1 = (1, 2, 4)^T, \mathbf{a}_2 = (3, 1, 7)^T$$

The reduced row echelon form involves one free variable and hence the nullspace will have dimension 1. Setting $x_3 = 1$, we get $x_1 = -2$ and $x_2 = 0$. Thus $(-2, 0, 1)^T$ is a basis for the null space.

Question 3. Let A be an $m \times n$ matrix with $m > n$. Let $\mathbf{b} \in \mathbb{R}^m$ and suppose that $\text{Null}(A) = \{\mathbf{0}\}$.

(a) Are the columns of A linearly independent? Do they span $\mathbb{R}^m$? Explain.

(b) How many solutions will the system $A\mathbf{x} = \mathbf{b}$ have if $\mathbf{b}$ is not in the column space of A? How many solutions will the system have if $\mathbf{b}$ is in the column space of A? Explain.

Solution

(a) Since $\text{Null}(A) = \{0\}$

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = 0$$

has only the trivial solution $\mathbf{x} = 0$, and hence $\mathbf{a}_1, \cdots, \mathbf{a}_n$ are linearly independent. The column vectors cannot span $\mathbb{R}^m$, since there are only n vectors and $n < m$.

(b) If $\mathbf{b}$ is not in the column space of A, then the system must be inconsistent and hence there will be no solutions. If $\mathbf{b}$ is in the column space of A, then the system will be consistent, so there will be at least one solution. By part (a), the column vectors are linearly independent, so there cannot be more than one solution. Thus, if $\mathbf{b}$ is in the column space of A, then the system will have exactly one solution.

Question 4. Let A be a 5×3 matrix of rank 3 and let $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ be a basis for $\mathbb{R}^3$.

(a) Show that $\text{Null}(A) = \{\mathbf{0}\}$.

(b) Show that if $\mathbf{y}_1 = A\mathbf{x}_1$, $\mathbf{y}_2 = A\mathbf{x}_2$, and $\mathbf{y}_3 = A\mathbf{x}_3$ then $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{y}_3$ are linearly independent.

(c) Do the vectors $\mathbf{y}_1$, $\mathbf{y}_2$, $\mathbf{y}_3$ from part (b) form a basis for $\mathbb{R}^5$? Explain.

Solution

(a) Since A is 5×3 with rank 3, its nullity is 0. Therefore $\text{Null}(A) = \{0\}$.

(b) If

$$c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3 = 0$$

then

$$c_1A\mathbf{x}_1 + c_2A\mathbf{x}_2 + c_3A\mathbf{x}_3 = 0$$

$$A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3) = 0$$

and it follows that $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3$ is in $\text{Null}(A)$. However, we know from

part (a) that $\text{Null}(A) = \{0\}$. Therefore

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = 0$$

Since $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly independent it follows that $c_1 = c_2 = c_3 = 0$ and hence $\mathbf{y}_1$, $\mathbf{y}_2$, $\mathbf{y}_3$ are linearly independent.
(c) Since $dim\mathbb{R}^5 = 5$ it takes 5 linearly independent vectors to span the vector space. The vectors $\mathbf{y}_1$, $\mathbf{y}_2$, $\mathbf{y}_3$ do not span $\mathbb{R}^5$ and hence cannot form a basis for $\mathbb{R}^5$.

Question 5. Let A be an $m \times n$ matrix.

(a) Show that if B is a nonsingular $m \times m$ matrix, then BA and A have the same null space and hence the same rank.

(b) Show that if C is a nonsingular $n \times n$ matrix, then AC and A have the same rank.

Solution

(a) If $x \in \text{Null}(A)$, then

$$BA\mathbf{x} = B0 = 0$$

and hence $x \in \text{Null}(BA)$. Thus $\text{Null}(A)$ is a subspace of $\text{Null}(BA)$. On the other hand, if $x \in \text{Null}(BA)$, then

$$B(A\mathbf{x}) = BA\mathbf{x} = 0$$

and hence $A\mathbf{x} \in \text{Null}(B)$. But $\text{Null}(B) = \{0\}$ since B is nonsingular. Therefore $A\mathbf{x} = 0$ and hence $\mathbf{x} \in \text{Null}(A)$. Thus BA and A have the same nullspace. It follows from the rank-nullity Theorem that

$$\begin{aligned}\text{rank}(A) &= n - \dim \text{Null}(A) \\ &= n - \dim \text{Null}(BA) \\ &= \text{rank}(BA)\end{aligned}$$

(b) By part (a), left multiplication by a nonsingular matrix does not alter the rank. Thus

$$\begin{aligned}\text{rank}(A) = \text{rank}(A^T) &= \text{rank}(C^T A^T) \\ &= \text{rank}((AC)^T) \\ &= \text{rank}(AC)\end{aligned}$$

Question 6. Let $\mathbf{x}$ and $\mathbf{y}$ be nonzero vectors in $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, and let $A = \mathbf{x}\mathbf{y}^T$.

(a) Show that $\{\mathbf{x}\}$ is a basis for the column space of A and that $\{\mathbf{y}^T\}$ is a basis for the row space of A.

(b) What is the dimension of Null(A)?

Solution

(a) Since

$$A = \mathbf{x}\mathbf{y}^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \mathbf{y}^T = \begin{bmatrix} x_1\mathbf{y}^T \\ x_2\mathbf{y}^T \\ \vdots \\ x_m\mathbf{y}^T \end{bmatrix}$$

the rows of A are all multiples of $\mathbf{y}^T$. Thus $\{\mathbf{y}^T\}$ is a basis for the row space of A. Since

$$\begin{aligned} A = \mathbf{x}\mathbf{y}^T &= \mathbf{x}(y_1, y_2, \cdots, y_n) \\ &= (y_1\mathbf{x}, y_2\mathbf{x}, \cdots, y_n\mathbf{x}) \end{aligned}$$

it follows that the columns of A are all multiples of $\mathbf{x}$ and hence $\{\mathbf{x}\}$ is a basis for the column space of A.

(b) Since A has rank 1, the nullity of A is $n - 1$.

Question 7. Prove the following properties for a general matrix A.

1. Null(A) = Null$(A^\top A)$.
2. rank(A) = rank$(A^\top A)$

Solution

(a) Since

Let $\mathbf{x} \in$ Null(A) where Null(A) is the null space of A.

$$A\mathbf{x} = \mathbf{0}$$

$$A^T A\mathbf{x} = \mathbf{0}$$

$$\mathbf{x} \in \text{Null}(A^T A)$$

Hence Null$(A) \subseteq$ Null$(A^T A)$. Let $\mathbf{x} \in$ Null$(A^T A)$

$$A^T A\mathbf{x} = \mathbf{0}$$

$$\mathbf{x}^T A^T A\mathbf{x} = \mathbf{0}$$

$$(A\mathbf{x})^T (A\mathbf{x}) = \mathbf{0}$$

$$A\mathbf{x} = \mathbf{0}$$

$$\mathbf{x} \in \text{Null}(A)$$

Hence $\text{Null}(A^TA) \subseteq \text{Null}(A)$.
Therefore, $\text{Null}(A) = \text{Null}(A^TA)$.
(b) Since

$$\text{Null}(A) = \text{Null}(A^TA)$$

$$dim(\text{Null}(A)) = dim(\text{Null}(A^TA))$$

$$dim(A) = dim(A^TA)$$

$$rank(A) = dim(C(A)) = dim(C(A^TA)) = rank(A^TA)$$